Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 23)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 A new lowland nature reserve is being created on former farmland next to an existing ancient woodland.

0 1 . 1 Explain why the **area** and the **shape** of the reserve both matter in its design.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 1 . 2 Explain the value of a biological corridor linking the new reserve to the ancient woodland.

[2 marks]

.....

.....

.....

.....

0 1 . 3 Explain how each of the following abiotic factors could be managed to increase the biodiversity of the reserve.

[4 marks]

Water levels .....

.....

Soil nutrient levels .....

.....

Topography .....

.....

Light levels .....

.....

Do not write outside the box

0 1 . 4 Much of the reserve will be kept as a **plagioclimax**. Explain what this means and how grazing is used to maintain one.

[3 marks]

0 1 . 5 Explain why the removal or culling of some species may be necessary on the reserve.

[3 marks]

15

0 2 . 1 A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate ☐

B The readings are accurate but not precise ☐

C The readings are both accurate and precise ☐

D The readings are reliable because the meter is electronic ☐

0 2 . 2 Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

Do not write outside the box

Do not write outside the box

.....

.....

.....

0 2 . 3 Explain why a preliminary study is carried out before the main investigation.

[2 marks]

.....

.....

.....

.....

5

0 3 . 1 Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

.....

.....

.....

.....

0 3 . 2 Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

0 3 . 3 Explain how auditory monitoring can be used to study bats.

[2 marks]

.....

.....

Do not write outside the box

.....

.....

0 3 . 4 Explain **three** assumptions of the mark-release-recapture method that may not be met.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

10

0 4 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 4 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

.....

.....

.....

.....

0 4 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

.....

.....

Do not write outside the box

5

0 5 . 1 Explain what is meant by **biomimetics** and describe **three** examples.

[4 marks]

0 5 . 2 Explain why many plants produce chemicals that are useful as human medicines, and give **two** examples.

[3 marks]

0 5 . 3 Explain why the discovery of medicines is a strong argument for conserving whole habitats rather than individual species.

[2 marks]

Do not write outside the box

0 5 . 4 Give **one** example of a wildlife species used in physiological research and the human health issue it helps to understand.

[1 mark]

10

0 6 . 1 Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

0 6 . 2 Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

0 6 . 3 Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]

.....

.....

.....

.....

10

0 7 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 7 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

5

Do not write outside the box

0 8 . 1 Explain why tropical rainforests have such high biodiversity.

[4 marks]

0 8 . 2 Explain why rainforest species may be particularly vulnerable to environmental change.

[2 marks]

0 8 . 3 Explain why rainforest plants and animals such as frogs are a rich source of medicines.

[2 marks]

0 8 . 4 Explain the importance of tropical rainforest for the hydrological cycle and for soils.

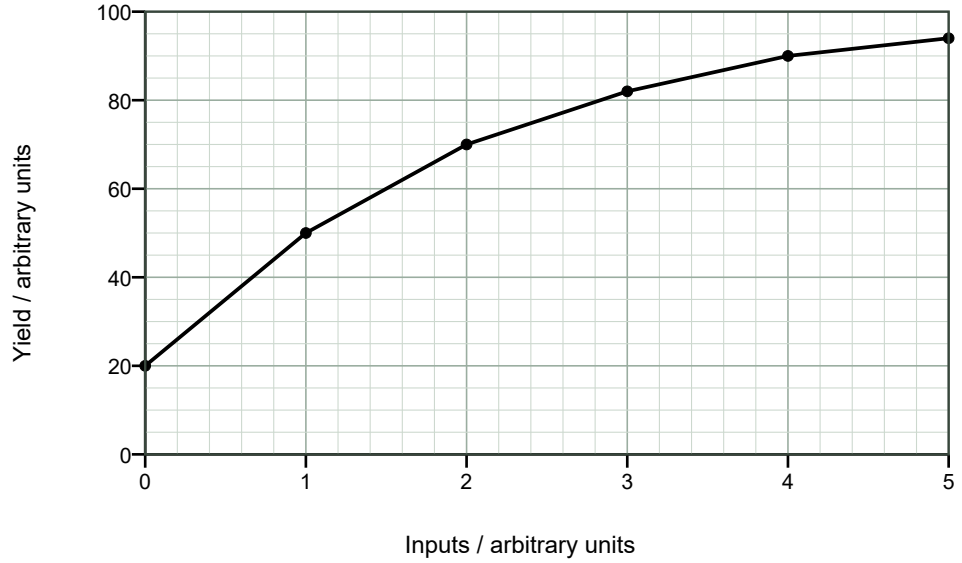
[2 marks]

Do not write outside the box

Do not write outside the box

0 9 Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1



0 9 . 1 Describe the relationship shown in Figure 1.

[2 marks]

.....

.....

.....

.....

0 9 . 2 Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]

Answer _____ units

0 9 . 3 Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

Do not write outside the box

0 9 . 4 Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

0 9 . 5 Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

10

1 0 . 1 State the Brundtland Commission definition of sustainable development.

[1 mark]

1 0 . 2 Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

Do not write outside the box

1 0 . 3 Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

1 0 . 4 Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

Do not write outside the box

Extra space

15

Write an essay on one of the following topics.

1 1 . 1 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

1 1 . 2 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS